

SRI HARSHA CHEDALLA

PLATFORM & AI ENGINEER | LINUX INTERNALS · DEEP LEARNING · FULL-STACK

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SUMMARY

Engineer who works across the full stack of intelligent systems — from the Linux kernel up to shipped AI products. Hardened production OS platforms at Samsung R&D (CVE patching, 20% performance gains, KSM-level memory optimization) and now build and deploy real-time deep-learning and RAG applications for live clients. Comfortable taking a system from a board on the desk to software people actually use, and motivated to build at startup speed.

EXPERIENCE

Platform Engineer Intern — Samsung R&D Institute Jan 2025 – Jul 2025 · Delhi

- Hardened the Linux platform by analyzing and **patching multiple CVEs** in kernel source code, strengthening the team's vulnerability-management and triage process.
- Improved OS performance by **20%** through targeted kernel configuration, memory-footprint reduction, and system-parameter tuning.
- Boosted memory efficiency by debugging and optimizing **Kernel Samepage Merging (KSM)**, increasing memory deduplication and resource utilization on constrained systems.
- Built custom Linux OS images using **Meson 4.9**, integrating updated kernel modules and drivers to meet specific platform requirements.
- Validated filesystem integrity, performance benchmarks, and reliability with **VDbench, VDLTP, and Auto Touch** test suites.
- Operated deep in Linux internals — kernel memory management, KSM implementation, and platform-level build pipelines — collaborating with platform teams to reproduce and resolve kernel issues and improve release stability.

Freelance Full-Stack & AI Developer Jul 2025 – Present

- **NAutomation Labs (AI Automation Agency)** — Designed, built, and deployed the agency's full web platform end-to-end: responsive front-end, service architecture, and a lead-capture funnel that translates complex AI-automation and AI-agent offerings into clear, conversion-focused pages. Owned UI/UX, development, and deployment, and iterated directly with the founder on positioning and copy.
- Shipped **5+ production client projects** across domains: **Sukhya Med** (medical appointment booking), **V7 Computers** (Shopify retail storefront), and **My Little Adventure** (travel site with integrated lead capture).
- Owned the complete delivery lifecycle — requirements, design, development, deployment, and client communication — consistently shipping working products to live users.

PROJECTS

SourcePath — Citation-Bound RAG for Legal Guidance Python · RAG · Vector Search · LLMs

- Built a Retrieval-Augmented Generation system that answers immigration-law questions with **every claim bound to its source citation**, sharply reducing hallucination and making answers auditable.
- Engineered the retrieval pipeline (document embeddings + vector search over a legal corpus) and grounded LLM responses in retrieved passages with inline source references; deployed as a live web app.

Real-Time Voice Emotion Detection PyTorch · Librosa · LSTM/RNN

- Built a real-time system that classifies human emotion (happy, sad, angry, etc.) from **live microphone audio** using an LSTM-based RNN architecture.
- Engineered an audio pipeline extracting **MFCC features** with Librosa and trained on the RAVDESS dataset, applying sequential modeling optimized for time-series audio.

EDUCATION

B.Tech, Electrical Engineering — National Institute of Technology, Durgapur (CGPA 8.28/10) 2021 – 2025

Class XII — Board of Intermediate Education, Andhra Pradesh (98.2%) · Class X (10.0 GPA) 2019 – 2021

TECHNICAL SKILLS

Languages	Python, C++, SQL, Bash / Shell Scripting
AI / ML	Deep Learning, NLP, PyTorch, LSTM / RNN, RAG, NumPy, Pandas
Systems	Linux Kernel Internals, Memory Management (KSM), Performance Tuning, Meson, Git
Web	HTML, CSS, Bootstrap, Shopify, Responsive Front-End
Foundations	Data Structures & Algorithms, Object-Oriented Programming